QUESTION ANALYSIS

EEE459

Compound Semiconductor and Hetero-Junction Devices

Question Analysis

Sessions 2020–21 *1 papers, 22 questions, 48 formulas*

INSIDE

• At a glance • Heat map • Priority list • 8 topic pages • Theory points • Derivations • Numericals • Diagrams to practise • Comparisons • Repeated questions • Styles • Study order

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Contents

1 At a glance **3**

2 Heat map **3**

3 Priority list **4**

4 Topic pages **4**

4.1 Band structure and materials 4
4.2 High-field transport 5
4.3 Alloys and strain 6
4.4 PN junction 7
4.5 Metal-semiconductor contacts 7
4.6 Heterojunctions 8
4.7 Bipolar transistors 9
4.8 JFETs 10

5 Theory points **11**

6 Derivations **12**

7 Numericals **13**

8 Diagrams to practise **14**

9 Comparisons **15**

10 Repeated questions **15**

11 Question style **16**

12 Study order **16**

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1. At a glance:

This analysis covers **1 paper** (2020-21), with 22 questions and 150 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

With one paper, no trend can be read. Use this as a map of the topics.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Metal-semiconductor contacts	31	21 %	5	Barrier lowering (14); Schottky junction (13); Ohmic contacts (4)
JFETs	29	19 %	4	Pinch-off (15); Transconductance (8); Channel profiles (6)
Alloys and strain	20	13 %	3	Lattice matching (9); Quaternary alloys (7); Isoelectronic traps (4)
Heterojunctions	20	13 %	2	Fields and potentials (12); Band diagrams (8)
Band structure and materials	17	11 %	3	Hole bands (6); Density of states (6); Materials (5)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2020-21	All
Band structure and materials	**17**	**17**
Hole bands	**6**	6
Density of states	**6**	6
Materials	**5**	5
High-field transport	**11**	**11**
Mobility	**6**	6
Gunn effect	**5**	5
Alloys and strain	**20**	**20**
Lattice matching	**9**	9
Quaternary alloys	**7**	7
Isoelectronic traps	**4**	4
PN junction	**5**	**5**
Graded junction	**5**	5
Metal-semiconductor contacts	**31**	**31**
Barrier lowering	**14**	14
Schottky junction	**13**	13
Ohmic contacts	**4**	4
Heterojunctions	**20**	**20**
Fields and potentials	**12**	12
Band diagrams	**8**	8
Bipolar transistors	**17**	**17**
Ebers-Moll	**9**	9
High-level injection	**8**	8
JFETs	**29**	**29**

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Topic and subtopic	2020-21	All
Pinch-off	**15**	15
Transconductance	**8**	8
Channel profiles	**6**	6

3. Priority list:

Must know: Metal-semiconductor contacts (31), JFETs (29), Alloys and strain (20), Heterojunctions (20), Band structure and materials (17), Bipolar transistors (17), High-field transport (11), PN junction (5)

Likely: none

Less often: none

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. Band structure and materials:

17 marks, 3 questions, in 1 of 1 papers. Book: Neamen Ch. 3; Sze Ch. 1.

Most often theory (2 of 3). Subtopics: Hole bands; Materials; Density of states.

Formulas:

Subtopic	Formula	Equation	Used in
Hole bands (6)	**Parabolic band**	$E = \frac{\hbar^2 k^2}{2m^*}$	20-21 1(c)
	Slope and curvature	$\frac{dE}{dk} = \frac{\hbar^2 k}{m^*}, \quad \frac{d^2E}{dk^2} = \frac{\hbar^2}{m^*}$	20-21 1(c)
	Si hole masses	$m_{hh,lh}[100] = \frac{m_0}{\gamma_1 \mp 2\gamma_2}, \quad m_{hh,lh}[111] = \frac{m_0}{\gamma_1 \mp 2\gamma_3}$	20-21 1(c)
Materials (5)	**Emission wavelength**	$\lambda\ (\mu\text{m}) = \frac{1.24}{E_g\ (\text{eV})}$	20-21 2(a)
Density of states (6)	**Density of states, 3D**	$g_{3D}(E) = \frac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E - E_c}$	20-21 3(b)
	Density of states, 2D	$g_{2D}(E) = \frac{m^*}{\pi\hbar^2}$ per subband	20-21 3(b)
	Density of states, 1D	$g_{1D}(E) \propto \frac{1}{\sqrt{E - E_n}}$	20-21 3(b)

Symbols:

E = energy from the band edge (J) ↩

$\hbar$ = reduced Planck constant (J · s) ↩

g = states per unit energy per unit volume

k = wave number (m^{-1}) ↩

λ = wavelength of a photon of energy E_g ↩

E_c = conduction-band edge ↩

Questions:

No.	Paper	Marks	Type	Question
Hole bands				
1	20-21 1(c)	6	derivation	Prove that along the [100] and [111] direction in k-space, for Si, the variation of electron energy with respect to wave number is higher for light holes than for heavy holes.
Materials				
2	20-21 2(a)	5	theory	Which semiconductors are most extensively used in near infra-red region? Explain the reasons behind it.
Density of states				
3	20-21 3(b)	6	theory	Explain how the density of states changes qualitatively as the dimensionality reduces from $3D \to 2D \to 1D$. Sketch $D(E)$ vs E for each case.

4.2. High-field transport:

11 marks, 2 questions, in 1 of 1 papers. Book: Sze Ch. 1, 10.

Most often theory (2 of 2). Subtopics: Mobility; Gunn effect.

Formulas:

Subtopic	Formula	Equation	Used in
Mobility (6)	**Mobility**	$\mu = \dfrac{e\,\tau}{m^*}$	20-21 1(a)
Gunn effect (5)	Drift velocity, low field	$v_d = \mu E$	20-21 1(b)
	Saturation velocity	$v_d \to v_{sat} \approx 10^7$ cm/s	20-21 1(b)
	Gunn oscillation	$f = \dfrac{v_d}{L}$	20-21 1(b)

Symbols:

μ = mobility ($\mathrm{cm}^2/\mathrm{V}\cdot\mathrm{s}$) ↩

τ = mean time between collisions (s) ↩

m^* = conductivity effective mass (kg) ↩

v_d = drift velocity (cm/s) ↩

E = electric field (V/cm) ↩

f = transit-time frequency (Hz) ↩

L = length of the GaAs sample (cm) ↩

Questions:

No.	Paper	Marks	Type	Question
Mobility				
1	20-21 1(a)	6	theory	Between Si and GaAs, which has higher electron mobility and why?
Gunn effect				
2	20-21 1(b)	5	theory	Discuss Velocity saturation and Gunn effect .

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4.3. Alloys and strain:

20 marks, 3 questions, in 1 of 1 papers. Book: Neamen Ch. 14.

Most often theory (2 of 3). Subtopics: Lattice matching; Isoelectronic traps; Quaternary alloys.

Formulas:

Subtopic	Formula	Equation	Used in
Lattice matching (9)	**Lattice misfit**	$f = \frac{a_{layer} - a_{sub}}{a_{sub}}$	20-21 2(b)
	Critical thickness, estimate	$h_c \sim \frac{a_{sub}}{2\lvert f\rvert}$	20-21 2(b)
Quaternary alloys (7)	**Vegard's law, ternary**	$a(A_xB_{1-x}C) = x\,a_{AC} + (1-x)\,a_{BC}$	20-21 2(d)
	Quaternary $In_xGa_{1-x}As_ySb_{1-y}$	$P = xy\,P_{InAs} + x(1-y)\,P_{InSb} + (1-x)y\,P_{GaAs} + (1-x)(1-y)\,P_{GaSb}$	20-21 2(d)

Symbols:

a_{layer} = layer lattice constant ↩

a_{sub} = substrate lattice constant ↩

a = lattice constant (Å) ↩

x = mole fraction ↩

P = lattice constant or band gap of the alloy, from the four binaries ↩

Questions:

No.	Paper	Marks	Type	Question
Lattice matching				
1	20-21 2(b)	9	theory	Why is lattice matching of semiconductor important for hetero-structured devices? Discuss the effect of strain and critical thickness in epitaxial growth.
Isoelectronic traps				
2	20-21 2(c)	4	theory	Discuss the effect of isoelectric trap in the ternary compound semiconductor $GaAs_{1-x}P_x$.
Quaternary alloys				
3	20-21 2(d)	7	numerical	Fig. 1 shows the Bandgap Vs. lattice constant for different binary compounds. Assume that a quaternary compound $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. Find the composition if the bandgap is 1 eV and lattice parameter is 5.93 Å.

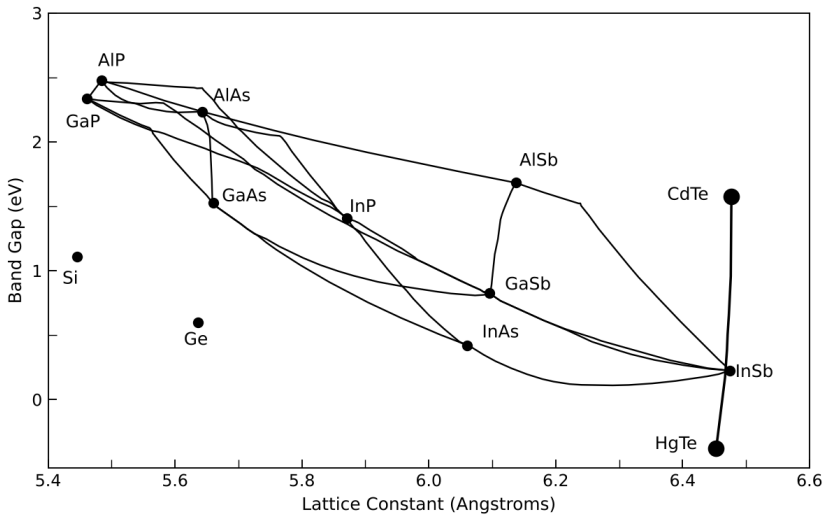


Figure 1: 20-21 2(d): Fig. 1: Band gap vs. lattice constant of binary compounds (redrawn from the paper)

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4.4. PN junction:

5 marks, 1 question, in 1 of 1 papers. Book: Neamen Ch. 7.

Most often theory (1 of 1). Subtopics: Graded junction.

Formulas:

Subtopic	Formula	Equation	Used in
Graded junction (5)	Net doping	$N_d - N_a = a\,x$	20-21 4(c)
	Depletion width	$W = \left[\frac{12\epsilon_s(V_{bi}+V_R)}{ea}\right]^{1/3}$	20-21 4(c)
	Junction capacitance	$C' = \left[\frac{ea\epsilon_s^2}{12(V_{bi}+V_R)}\right]^{1/3}$	20-21 4(c)

Symbols:

a = grading constant (cm^{-4}) ↩ W = total depletion width (cm) ↩

Questions:

No.	Paper	Marks	Type	Question
Graded junction				
1	20-21 4(c)	5	theory	Describe how a linearly graded junction is formed.

4.5. Metal-semiconductor contacts:

31 marks, 5 questions, in 1 of 1 papers. Book: Neamen Ch. 9; Sze Ch. 3.

Most often theory (3 of 5). Subtopics: Schottky junction; Barrier lowering; Ohmic contacts.

Formulas:

Subtopic	Formula	Equation	Used in
Schottky junction (13)	**Barrier height (ideal)**	$\phi_{B0} = \phi_m - \chi$	20-21 3(c)
	Built-in potential	$V_{bi} = \phi_{B0} - \phi_n,\ \phi_n = \frac{kT}{e}\ln\frac{N_c}{N_d}$	20-21 3(c)
	Depletion width	$x_n = \left[\frac{2\epsilon_s(V_{bi}+V_R)}{eN_d}\right]^{1/2}$	20-21 3(c)
	Maximum field	$E_{max} = \frac{eN_dx_n}{\epsilon_s} = \frac{2(V_{bi}+V_R)}{x_n}$	20-21 3(c)
	Junction capacitance	$C' = \frac{\epsilon_s}{x_n} = \left[\frac{e\epsilon_sN_d}{2(V_{bi}+V_R)}\right]^{1/2}$	20-21 3(c)
Barrier lowering (14)	**Image-force lowering**	$\Delta\phi = \sqrt{\frac{eE}{4\pi\epsilon_s}}$	20-21 3(d)
	Position of the peak	$x_m = \sqrt{\frac{e}{16\pi\epsilon_sE}}$	20-21 3(d)
	Barrier with interface states	$\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)\left(\frac{E_g}{e} - \phi_0\right)$	20-21 4(b)
Ohmic contacts (4)	Ohmic contact (ideal)	$\phi_m < \phi_s$ (n-type), $\phi_m > \phi_s$ (p-type)	20-21 5(d)

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Symbols:

ϕ_m = metal work function (V) ↩

χ = electron affinity (V) ↩

ϕ_n = depth of E_F below E_c (V) ↩

N_c = conduction-band density of states (cm^{-3}) ↩

V_R = reverse bias (V) ↩

C' = capacitance per unit area (F/cm^2) ↩

$\Delta\phi$ = barrier lowering (V) ↩

E = field at the interface (V/cm)

x_m = distance of the barrier maximum from the metal (cm) ↩

$\gamma = 1$ without interface states, 0 when the Fermi level is pinned ↩

ϕ_0 = neutral level of the interface states above E_v (V) ↩

ϕ_s = semiconductor work function (V) ↩

Questions:

No.	Paper	Marks	Type	Question
Schottky junction				
1	20-21 3(a)	5	theory	Draw the ideal energy band diagram of a metal semiconductor junction under: (i) Reverse bias condition (ii) Forward bias condition
2	20-21 3(c)	8	numerical	Consider a contact between tungsten and n-type silicon doped to $N_d = 1 \times 10^{16}$ cm^{-3} at $T = 300$ K. Determine the (i) theoretical barrier height (ii) built-in potential barrier (iii) maximum electric field in the metal-semiconductor diode for zero applied bias, and (iv) the junction capacitance. Given that, the metal work function for tungsten (W) is 4.55 V and the electron affinity for silicon is 4.01 V. Assume $N_c = 2.8 \times 10^{19}$ cm^{-3}.
Barrier lowering				
3	20-21 3(d)	6	numerical	Consider a gallium arsenide metal-semiconductor contact in which the electric field in the semiconductor is assumed to be $E = 6.8 \times 10^4$ V cm^{-1}. Calculate the Schottky barrier lowering (due to image force induced lowering) and the position of the maximum barrier height.
4	20-21 4(b)	8	theory	Discuss the following non-ideal effects on barrier height of a metal-semiconductor heterojunction: (i) Image-force-induced lowering (ii) Interface states
Ohmic contacts				
5	20-21 5(d)	4	theory	Sketch the ideal energy band diagram of a metal semiconductor junction in which $\Phi_m < \Phi_s$. Explain why this is an ohmic contact.

4.6. Heterojunctions:

20 marks, 2 questions, in 1 of 1 papers. Book: Neamen Ch. 9.

Most often derivation (2 of 2). Subtopics: Band diagrams; Fields and potentials.

Formulas:

Subtopic	Formula	Equation	Used in
Band diagrams (8)	AlGaAs band gap	$E_g(Al_xGa_{1-x}As) = 1.424 + 1.247x$ eV	20-21 1(d)
	Band offsets (Anderson)	$\Delta E_c = e(\chi_n - \chi_w),\ \Delta E_g = \Delta E_c + \Delta E_v$	20-21 1(d)
	Hole concentration	$p = N_v \exp\left[-\frac{E_F - E_v}{kT}\right]$	20-21 1(d)
	Built-in potential, pN	$eV_{bi} = -\Delta E_v + kT \ln\left(\frac{p_P\, N_{vN}}{p_N\, N_{vP}}\right)$	20-21 1(d)
Fields and potentials (12)	Field, N side	$E_1(x) = \frac{eN_{d1}}{\epsilon_1}(x + x_n)$	20-21 4(a)
	Field, P side	$E_2(x) = \frac{eN_{a2}}{\epsilon_2}(x_p - x)$	20-21 4(a)

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Subtopic	Formula	Equation	Used in
	Interface condition	$\epsilon_1 E_1(0) = \epsilon_2 E_2(0) \Rightarrow N_{d1} x_n = N_{a2} x_p$	20-21 4(a)
	Potential split	$V_{bi1} = \frac{eN_{d1}x_n^2}{2\epsilon_1},\ V_{bi2} = \frac{eN_{a2}x_p^2}{2\epsilon_2},\ V_{bi} = V_{bi1} + V_{bi2}$	20-21 4(a)
	Depletion width, N side	$x_n = \left[\frac{2\epsilon_1\epsilon_2 N_{a2} V_{bi}}{eN_{d1}(\epsilon_1 N_{d1} + \epsilon_2 N_{a2})}\right]^{1/2}$	20-21 4(a)

Symbols:

χ_n, χ_w = electron affinities of the narrow- and wide-gap materials (V) ↩

ΔE_c, ΔE_v = conduction- and valence-band offsets (eV) ↩

N_v = valence-band effective density of states (cm^{-3}) ↩

p_P, p_N = hole concentrations in the P and N regions ↩

N_{vP}, N_{vN} = their valence-band densities of states ↩

N_{d1} = donors ↩

ϵ_1 = permittivity of side 1 ↩

N_{a2} = acceptors ↩

ϵ_2 = permittivity of side 2 ↩

V_{bi1}, V_{bi2} = parts of V_{bi} dropped on each side ↩

Questions:

No.	Paper	Marks	Type	Question
Band diagrams				
1	20-21 1(d)	8	derivation	For a $GaAs - Al_xGa_{1-x}As$ heterojunction, determine the following: (i) Energy-Band diagram: Case-1: Before contact, Case-2: After contact (ii) Show that the built-in potential (using hole concentration) $eV_{bi} = -\Delta E_v + kT \ln\left(\frac{P_P N_{VN}}{P_N N_{VP}}\right)$. Given, GaAs is doped P type, $Al_xGa_{1-x}As$ is doped N^+ type. Assume: $x = 0.25$ & $\Delta E_c = \frac{1}{2}\Delta E_g$ P_p = hole concentration in P type region, P_N = hole concentration in N type region, N_{VN} and N_{VP} are density of states of electron and hole respectively.
Fields and potentials				
2	20-21 4(a)	12	derivation	A narrow bandgap N-type and a wide bandgap P-type material are used to form a heterojunction. If they are straddling type, then draw the energy band diagram of the heterojunction before contact and after contact at thermal equilibrium. With necessary assumption, derive expression for electric field and electrostatic potential in both regions.

4.7. Bipolar transistors:

17 marks, 2 questions, in 1 of 1 papers. Book: Neamen Ch. 12; Sze Ch. 5.

Most often derivation (1 of 2). Subtopics: Ebers-Moll; High-level injection.

Formulas:

Subtopic	Formula	Equation	Used in
Ebers-Moll (9)	**Ebers–Moll currents**	$I_C = \alpha_F I_F - I_R,\ I_E = I_F - \alpha_R I_R,\ I_B = I_E - I_C$	20-21 6(a)
	Diode currents	$I_F = I_{ES}\left[e^{V_{BE}/V_t} - 1\right],\ I_R = I_{CS}\left[e^{V_{BC}/V_t} - 1\right]$	20-21 6(a)
	Reciprocity	$\alpha_F I_{ES} = \alpha_R I_{CS}$	20-21 6(a)
	Saturation voltage	$V_{CE}(sat) = V_t \ln\left[\frac{I_C(1-\alpha_R) + I_B}{\alpha_F I_B - (1-\alpha_F) I_C} \cdot \frac{\alpha_F}{\alpha_R}\right]$	20-21 6(a)

Symbols:

α_F, α_R = forward and reverse common-base gains ↩

I_{ES}, I_{CS} = reverse saturation currents of the B–E and B–C junctions (A) ↩

Questions:

No.	Paper	Marks	Type	Question
Ebers-Moll				
1	20-21 6(a)	9	derivation	Draw the equivalent circuit for basic Ebers-Moll model as used in BJT. Also derive the expression of $V_{CE}(sat)$ from this model.
High-level injection				
2	20-21 6(b)	8	theory	Explain high level injection and current crowding effects in a BJT with necessary figures. How can the problem of current crowding be solved?

4.8. JFETs:

29 marks, 4 questions, in 1 of 1 papers. Book: Neamen Ch. 13; Sze Ch. 7.

Most often numerical (2 of 4). Subtopics: Channel profiles; Pinch-off; Transconductance.

Formulas:

Subtopic	Formula	Equation	Used in
Channel profiles (6)	Saturation drain voltage	$V_{DS}(sat) = V_{p0} - (V_{bi} - V_{GS})$	20-21 5(a)
Pinch-off (15)	pn built-in potential	$V_{bi} = V_t \ln \dfrac{N_a N_d}{n_i^2}$	20-21 5(c)
	Space-charge width	$h = \left[\dfrac{2\epsilon_s(V_{bi} - V_{GS})}{eN_d}\right]^{1/2}$	20-21 5(b)
	Internal pinch-off voltage	$V_{p0} = \dfrac{ea^2 N_d}{2\epsilon_s}$	20-21 5(b)
	Pinch-off voltage	$V_p = V_{bi} - V_{p0}$	20-21 5(b)
Transconductance (8)	**Pinch-off current**	$I_{P1} = \dfrac{\mu_n (eN_d)^2 W a^3}{6\epsilon_s L}$	20-21 6(c)
	Saturation current	$I_{D1}(sat) = I_{P1}\left[1 - 3\left(\dfrac{V_{bi} - V_{GS}}{V_{p0}}\right)\left(1 - \dfrac{2}{3}\sqrt{\dfrac{V_{bi} - V_{GS}}{V_{p0}}}\right)\right]$	20-21 6(c)
	Transconductance	$g_{ms} = \dfrac{3I_{P1}}{V_{p0}}\left(1 - \sqrt{\dfrac{V_{bi} - V_{GS}}{V_{p0}}}\right)$	20-21 6(c)

Symbols:

N_a = gate acceptors ↩

N_d = channel donors (cm^{-3}) ↩

h = depletion width into the channel (cm) ↩

V_{GS} = gate–source voltage (V) ↩

a = metallurgical channel thickness (cm)

W = channel width

L = channel length (cm)

μ_n = electron mobility ($\text{cm}^2/\text{V}\cdot\text{s}$) ↩

g_{ms} = saturation transconductance (S) ↩

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Questions:

No.	Paper	Marks	Type	Question
Channel profiles				
1	20-21 5(a)	6	theory	Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where V_p, V_g, and V_d are pinch-off, gate, and drain voltage respectively) (i) $V_g > V_p$ and $V_d < V_{d,sat}$ (ii) $V_g > V_p$ and $V_d = V_{d,sat}$ (iii) $V_g > V_p$ and $V_d > V_{d,sat}$
Pinch-off				
2	20-21 5(b)	8	derivation	Derive the formula of internal pinch-off voltage in an n-channel JFET.
3	20-21 5(c)	7	numerical	Consider a GaAs n-channel pn JFET at $T = 300$ K with $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 3 \times 10^{15}$ cm^{-3}, $\epsilon_s = 13.1$, and $a = 0.70$ μm. Determine the forward-bias gate voltage required to open a channel region that is 0.10 μm thick with zero drain voltage.
Transconductance				
4	20-21 6(c)	8	numerical	Consider a silicon n-channel JFET at $T = 300$ K with the following parameters: $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 1 \times 10^{16}$ cm^{-3}, $a = 0.75$ μm, $L = 10$ μm, $W = 30$ μm, and $\mu_n = 1000$ cm^2/(V s) (i) Calculate the maximum current in an n-channel JFET. (ii) Determine the maximum transconductance of an n-channel depletion mode JFET biased in the saturation region. $I_{P1} = 0.522$ mA, $V_{bi} = 0.814$ V, and $V_{po} = 4.35$ V.

5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

No.	Paper	Marks	Subtopic	Points to cover
Band structure and materials				
1	20-21 2(a)	5	Materials	Near-IR (0.75 to 2 μm) uses III-V compounds: GaAs/AlGaAs (0.8 to 0.9 μm), InGaAsP and InGaAs on InP (1.3 and 1.55 μm, the fibre windows), Ge or InGaAs detectors. Reasons: band gaps 0.6 to 1.65 eV give $\lambda = 1.24/E_g$ in the near-IR; direct gaps give efficient emission; alloys tune the gap while lattice-matched to InP or GaAs.
2	20-21 3(b)	6	Density of states	$D(E)$ counts states per unit energy. 3D: $D \propto \sqrt{E - E_c}$. 2D quantum well: constant $m^*/\pi\hbar^2$ per subband, a staircase touching the 3D curve. 1D wire: $D \propto 1/\sqrt{E - E_n}$ spikes at each subband edge. 0D dot: delta functions. Confinement squeezes states into sharper features.
High-field transport				
3	20-21 1(a)	6	Mobility	GaAs: $\mu_n = 8500$ against 1350 cm^2/V s for Si. $\mu = e\tau/m^*$: GaAs has one sharply curved direct Γ valley with $m^* = 0.067\,m_0$; Si has six flat X valleys (conductivity mass $0.26\,m_0$) and intervalley scattering among them; GaAs also has weaker phonon scattering in the Γ valley.
4	20-21 1(b)	5	Gunn effect	Velocity saturation: hot carriers lose their extra energy to optical phonons, so v_d levels off near 10^7 cm/s (Si above about 3×10^4 V/cm). Gunn effect: in n-GaAs above about 3.2 kV/cm electrons transfer from the light Γ valley to the heavy L valley, v_d falls (negative differential mobility), and high-field domains form and travel, giving microwave current pulses.
Alloys and strain				
5	20-21 2(b)	9	Lattice matching	Matching ($f = (a_L - a_S)/a_S \approx 0$) avoids misfit and threading dislocations, which are non-radiative recombination and scattering centres. A thin mismatched layer grows strained (pseudomorphic, tetragonal distortion) up to a critical thickness h_c (Matthews–Blakeslee), beyond which dislocations relax it; strain also shifts band edges and can be used on purpose.

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No.	Paper	Marks	Subtopic	Points to cover
6	20-21 2(c)	4	Isoelectronic traps	$GaAs_{1-x}P_x$ is direct for $x < 0.45$ and indirect above. Nitrogen on a P site is isoelectronic and forms a trap just below E_c; the electron is localised in space, so spread in k, and can recombine with a hole without a phonon; this makes indirect GaAsP:N (red-orange, yellow, green) LEDs efficient.
PN junction				
7	20-21 4(c)	5	Graded junction	Diffuse acceptors into a uniformly doped n-type wafer (or grade the doping during epitaxy); near the depth where $N_a = N_d$ the net doping is $a(x - x_j)$: linear. Then $\rho = eax$, the field is parabolic, $W = [12\epsilon_s(V_{bi} + V_R)/(ea)]^{1/3}$ and $C' \propto (V_{bi} + V_R)^{-1/3}$.
Metal-semiconductor contacts				
8	20-21 3(a)	5	Schottky junction	Ideal metal/n-type contact: the barrier from the metal stays $e\phi_{B0} = e(\phi_m - \chi)$. Reverse bias raises the barrier seen from the semiconductor to $e(V_{bi} + V_R)$, widens the depletion layer, tiny current. Forward bias lowers it to $e(V_{bi} - V_a)$, narrows the layer, electrons flow into the metal.
9	20-21 4(b)	8	Barrier lowering	(i) The image charge of an electron lowers the barrier by $\Delta\phi = \sqrt{eE/4\pi\epsilon_s}$ at $x_m = \sqrt{e/16\pi\epsilon_s E}$; E grows with reverse bias, so the reverse current rises with V_R. (ii) Interface states with a neutral level $e\phi_0$ pin the surface Fermi level, so the barrier depends only weakly on the metal work function (Bardeen limit).
10	20-21 5(d)	4	Ohmic contacts	With $\phi_m < \phi_s$ on n-type, electrons flow from the metal into the semiconductor and form an accumulation layer; the bands bend down by $e(\phi_s - \phi_m)$; electrons meet no barrier either way, so the I-V is linear with low resistance: an ohmic contact.
Bipolar transistors				
11	20-21 6(b)	8	High-level injection	High-level injection: injected electrons in the base approach N_B, extra holes come in for neutrality, effective base doping rises, injection efficiency falls, $I_C \propto e^{V_{BE}/2V_t}$, β falls at high current. Current crowding: the lateral base current's IR drop makes V_{BE} largest at the emitter edges, so current crowds there (hot spots). Cure: narrow interdigitated emitter fingers, more base contacts, ballast resistors.
JFETs				
12	20-21 5(a)	6	Channel profiles	The gate–channel reverse bias grows from source to drain, so the depletion layer is a wedge, thickest at the drain. (i) $V_d < V_{d,sat}$: channel open throughout, linear region. (ii) $V_d = V_{d,sat}$: the depletion just touches the bottom at the drain (pinch-off). (iii) $V_d > V_{d,sat}$: the pinched-off part grows towards the source, current saturates.

6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Band structure and materials				
1	20-21 1(c)	6	Hole bands	Parabolic band $E = E_v - \hbar^2k^2/2m^*$; slope $\hbar^2k/m^*$; ratio of slopes at the same k is m_{hh}/m_{lh}; Luttinger masses along [100] (γ_2) and [111] (γ_3); ratios 1.45 and 5.30, both above 1.
Heterojunctions				
2	20-21 1(d)	8	Band diagrams	E_g of AlGaAs from x, $\Delta E_c = \Delta E_v = 0.156$ eV; band diagrams before (common vacuum level) and after contact (flat E_F, spike and step); hole concentration on each side; solve for E_F; $eV_{bi} = E_{FN} - E_{FP}$; put $E_{vN} - E_{vP} = -\Delta E_v$.

No.	Paper	Marks	Derivation	Outline
3	20-21 4(a)	12	Fields and potentials	$\rho = +eN_{d1}$, $-eN_{a2}$; Poisson in each material with its own ϵ; fields zero at the edges; $\epsilon_1 E_1(0) = \epsilon_2 E_2(0)$ gives $N_{d1}x_n = N_{a2}x_p$; integrate again with ϕ continuous; $V_{bi} = V_{bi1} + V_{bi2}$; solve for x_n, x_p.
Bipolar transistors				
4	20-21 6(a)	9	Ebers-Moll	Two diodes I_F, I_R with sources $\alpha_F I_F$, $\alpha_R I_R$; KCL gives I_C, I_E, I_B; saturation: drop the -1 terms, V_{BE} and V_{BC} as logarithms; solve for I_F, I_R from I_C, I_B; $V_{CE} = V_{BE} - V_{BC}$; reciprocity $I_{CS}/I_{ES} = \alpha_F/\alpha_R$.
JFETs				
5	20-21 5(b)	8	Pinch-off	Poisson with $\rho = +eN_d$ in the depletion layer; $E = 0$ at $y = h$; potential $eN_d h^2/2\epsilon_s$; equals $V_{bi} - V_{gs}$; gives h; pinch-off at $h = a$; $V_{p0} = ea^2 N_d/2\epsilon_s$, $V_p = V_{bi} - V_{p0}$.

7. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

No.	Paper	Marks	Subtopic	Method	Answer
Alloys and strain					
1	20-21 2(d)	7	Quaternary alloys	Read the corner binaries from Fig. 1; Vegard's law (bilinear interpolation) for $a(x,y)$ and $E_g(x,y)$ of $\mathrm{In}_x\mathrm{Ga}_{1-x}\mathrm{As}_y\mathrm{Sb}_{1-y}$; set $a = 5.93$ Å and $E_g = 1$ eV; solve the two equations.	$x \approx 0.24$, $y \approx 0.60$ (straight-line interpolation; with the bowing printed in Fig. 1 no alloy has both values)
Metal-semiconductor contacts					
2	20-21 3(c)	8	Schottky junction	$\phi_{B0} = \phi_m - \chi$; $\phi_n = V_t \ln(N_c/N_d)$; $V_{bi} = \phi_{B0} - \phi_n$; $x_n = [2\epsilon_s V_{bi}/(eN_d)]^{1/2}$; $\lvert E_{max}\rvert = eN_d x_n/\epsilon_s$; $C' = \epsilon_s/x_n$.	$\phi_{B0} = 0.54$ V; $V_{bi} = 0.334$ V; $\lvert E_{max}\rvert = 3.22 \times 10^4$ V/cm; $C' = 4.98 \times 10^{-8}$ F/cm^2
3	20-21 3(d)	6	Barrier lowering	Image-force lowering $\Delta\phi = \sqrt{eE/(4\pi\epsilon_s)}$; position $x_m = \sqrt{e/(16\pi\epsilon_s E)}$ with ϵ_s of GaAs.	$\Delta\phi = 27.3$ mV at $x_m = 2.01$ nm
JFETs					
4	20-21 5(c)	7	Pinch-off	$V_{bi} = V_t \ln(N_a N_d/n_i^2)$; $V_{p0} = ea^2 N_d/2\epsilon_s$; $V_p = V_{bi} - V_{p0}$; for an open channel $a - h = 0.10$ μm, $h = 0.60$ μm, $V_{bi} - V_{gs} = eN_d h^2/2\epsilon_s$, solve for V_{gs}.	$V_{bi} = 1.250$ V; $V_{p0} = 1.016$ V; $V_p = +0.235$ V; $V_{gs} \approx +0.50$ V (forward)
5	20-21 6(c)	8	Transconductance	$I_{P1} = \mu_n (eN_d)^2 W a^3/(6\epsilon_s L)$; V_{bi} and V_{p0}; $I_D(sat)$ at $V_{GS} = 0$ from the JFET saturation equation; $g_{ms} = (3I_{P1}/V_{p0})(1 - \sqrt{(V_{bi} - V_{GS})/V_{p0}})$ at $V_{GS} = 0$.	$I_{P1} = 0.522$ mA ($I_{D1}(sat) = 0.3135$ mA at $V_{GS} = 0$); $g_{ms}(max) = 0.204$ mS

8. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

No.	Paper	Kind	Diagram
Band structure and materials			
1	20-21 1(c)	Graph	E-k of the Si heavy- and light-hole bands along [111] and [100]
2	20-21 2(a)	Graph	fibre loss windows and the wavelength ranges of the materials
3	20-21 3(b)	Diagram	counting allowed k points in 1D, 2D and 3D
4	20-21 3(b)	Graph	density of states against energy in 3D, 2D, 1D and 0D
High-field transport			
5	20-21 1(a)	Diagram	electron drift with and without a field
6	20-21 1(a)	Diagram	simplified E-k diagrams of Si and GaAs
7	20-21 1(b)	Graph	drift velocity against field for Si and GaAs
8	20-21 1(b)	Diagram	the two GaAs valleys and electron transfer
9	20-21 1(b)	Diagram	a travelling domain in a Gunn diode and the current pulses
Alloys and strain			
10	20-21 2(b)	Diagram	matched, strained and relaxed layers on a substrate
11	20-21 2(b)	Graph	critical thickness against misfit
12	20-21 2(c)	Graph	GaAsP band gap against x with direct and indirect branches
13	20-21 2(d)	Graph	InGaAsSb region of Fig. 1 with the target point
PN junction			
14	20-21 4(c)	Diagram	graded junction by diffusion
15	20-21 4(c)	Graph	diffused acceptor profile and net doping
16	20-21 4(c)	Graph	linear charge, parabolic field
Metal-semiconductor contacts			
17	20-21 3(a)	Diagram	metal–n-semiconductor band diagrams at zero, reverse and forward bias
18	20-21 3(c)	Diagram	W and n-Si before and after contact
19	20-21 3(c)	Graph	space charge and field in the depletion region
20	20-21 3(d)	Diagram	electron and its image charge, and the lowered barrier
21	20-21 4(b)	Graph	barrier with image-force lowering
22	20-21 4(b)	Diagram	metal–semiconductor with an interfacial layer and interface states
23	20-21 4(b)	Graph	barrier height against metal work function
24	20-21 5(d)	Diagram	metal and n-semiconductor band diagrams for $\phi_m < \phi_s$
25	20-21 5(d)	Graph	ohmic and rectifying I-V curves
Heterojunctions			
26	20-21 1(d)	Diagram	the three heterojunction band alignments
27	20-21 1(d)	Diagram	p-GaAs and N-AlGaAs before contact
28	20-21 1(d)	Diagram	the heterojunction after contact in equilibrium
29	20-21 4(a)	Diagram	narrow-gap N and wide-gap P before contact
30	20-21 4(a)	Diagram	equilibrium band diagram of the straddling nP junction
31	20-21 4(a)	Graph	charge, field and potential with the field jump at $x = 0$
Bipolar transistors			
32	20-21 6(a)	Diagram	npn regions and currents
33	20-21 6(a)	Circuit	Ebers-Moll equivalent circuit
34	20-21 6(a)	Graph	output curves with the saturation region, and $V_{CE}(sat)$ against I_B
35	20-21 6(b)	Graph	Gummel plot with high-level injection
36	20-21 6(b)	Diagram	emitter current crowding under the emitter
37	20-21 6(b)	Diagram	interdigitated emitter-base layout
JFETs			
38	20-21 5(a)	Diagram	channel space-charge profiles in the three regions

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No.	Paper	Kind	Diagram
39	20-21 5(a)	Graph	drain current against drain voltage with the three points
40	20-21 5(b)	Diagram	channel geometry of single- and two-gate JFETs
41	20-21 5(b)	Graph	charge, field and potential across the gate junction
42	20-21 5(c)	Graph	depletion width against gate voltage
43	20-21 6(c)	Graph	transfer curve and transconductance

9. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

Pair	Key difference	Asked in
Band structure and materials		
3D vs 2D vs 1D density of states	$\sqrt{E}$ vs steps vs $1/\sqrt{E}$ spikes	20-21 3(b)
High-field transport		
Si vs GaAs electrons	indirect gap, six flat X valleys, $m^* = 0.26\,m_0$ vs direct gap, one curved Γ valley, $m^* = 0.067\,m_0$, 6 times the mobility	20-21 1(a)
Velocity saturation vs Gunn effect	velocity levels off (Si) vs velocity falls after a peak, negative mobility, oscillation (GaAs)	20-21 1(b)
Alloys and strain		
Strained vs relaxed layer	thinner than h_c, no dislocations, distorted cell vs thicker, misfit dislocations, own lattice constant	20-21 2(b)
PN junction		
Abrupt vs linearly graded junction	step doping, $W \propto V^{1/2}$ vs linear doping, $W \propto V^{1/3}$	20-21 4(c)
Metal-semiconductor contacts		
Reverse vs forward biased Schottky junction	barrier $V_{bi} + V_R$, wider depletion, small current vs barrier $V_{bi} - V_a$, narrower, large electron current	20-21 3(a)
Schottky limit vs Bardeen limit	barrier follows $\phi_m - \chi$ vs barrier pinned by interface states	20-21 4(b)
Ohmic vs rectifying contact	$\phi_m < \phi_s$ (n-type), no barrier, linear I-V vs $\phi_m > \phi_s$, Schottky barrier, diode I-V	20-21 5(d)
Heterojunctions		
Straddling vs staggered vs broken-gap heterojunction	one gap inside the other vs offsets in the same direction vs the gaps do not overlap	20-21 1(d)
Bipolar transistors		
High-level injection vs current crowding	base charge rises, β falls vs lateral IR drop, current concentrates at emitter edges	20-21 6(b)

10. Repeated questions:

No repeats among these papers.

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11. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical
Band structure and materials	2	1	·
High-field transport	2	·	·
Alloys and strain	2	·	1
PN junction	1	·	·
Metal-semiconductor contacts	3	·	2
Heterojunctions	·	2	·
Bipolar transistors	1	1	·
JFETs	1	1	2

12. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Metal-semiconductor contacts	31	21 %
2	JFETs	29	40 %
3	Alloys and strain	20	53 %
4	Heterojunctions	20	67 %
5	Band structure and materials	17	78 %
6	Bipolar transistors	17	89 %
7	High-field transport	11	97 %
8	PN junction	5	100 %

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